Risk factors and seroprevalence of hepatitis C virus antibody among blood donors in Lagos

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Summary
Aims and objectives: *This study was undertaken (i) to determine the seroprevalence of antibodies to Hepatitis C virus (anti –HCV) among blood donors (ii) to document the incidence of known risk factors for HCV infection among blood donors.*
Patients and Methods: *This is a cross sectional prospective study among apparently normal blood donors. Subjects were recruited from three different hospitals in Lagos metropolis. All recruited donors were evaluated for HCV infection-associated risk factors by questionnaire interviews. Sera samples from recruited donors were tested for anti-HCV using third generation Murex (Murex Biotech, South Africa) and fourth generation Dialab Enzyme Linked Immunosorbent Assay (ELISA) kits (Dialab. Austria).*
Results: *A total of three hundred and thirty four blood donors were screened, of which seven (2.1 %) were positive for anti-HCV.*
The blood donors comprised 15 (4.5 %) females and 319 (95.5 %) males.
There was an association between anti-HCV positivity and history of multiple sex partners and previous sexually transmitted infections (X2-15.9; p<0.05)
Majority of blood donors were family replacement 317 (94.9 %) with anti-HCV prevalence of 2.2 % (7/317) while 5.1 % (17/334) were voluntary non remunerated with anti-HCV prevalence of 0% (p>0.05)
Conclusion: *Prevalence of anti-HCV among blood donors in Lagos (2.1%) is low as in most previous reports from Nigeria and some other parts of Africa*

Key words: hepatitis C virus antibody; prevalence; risk factors; blood donors.

Introduction

Hepatitis C virus (HCV) causes acute and chronic post-transfusion hepatitis infection, and also morbidity and mortality worldwide [1]. HCV infection has an asymptomatic clinical picture in most cases with 80% of the infected subjects developing chronic hepatitis, cirrhosis of the liver or hepatocellular carcinoma[1].

Transmission of HCV is predominantly by parenteral route, through blood and blood products. It could also be transmitted through the sharing of syringes, cannulae or by other very small injuries involving blood-blood contact[2].

The risk of infection from HCV- contaminated blood is in excess of 75 %.[1]

Similarly, blood products for example factor VIII, are also a major source of HCV infection and reports of the prevalence of infection in haemophiliacs have varied from 30-80 %.[1] Other high risk groups for infection with HCV are intravenous drug users, renal haemodialysis patients and recipients of multiple blood or blood products transfusions[3]

Sexual transmission of HCV infection may occur, while vertical transmission may be responsible for a few infections[1].

This study was undertaken to determine the seroprevalence of hepatitis C antibody and to document the incidence of known risk factors for HCVinfection among blood donors.

Materials & Methods:

After ethical approval, apparently healthy blood donors were recruited from the Blood Donors' clinics of three hospitals i.e. Lagos University Teaching Hospital, Lagos State University Teaching Hospital and Lagos Island General Hospital.

A structured questionnaire was used to collect demographic data and medical history from consecutive donors. Blood donors with positive history of previous blood transfusion were excluded from this study.

Anaemia was excluded in every prospective donor by haemoglobin estimation using an electronic haemoglobinometry method; the Haemocue technique. All prospective donors whose haemoglobin level was below 12.5g/l were excluded.

Five milliliters of blood was collected into universal bottle from each donor after informed verbal consent. Blood samples were allowed to clot and retract before separation of sera which was stored at -30^{0}C before being analysed.

Screening of sera samples for anti-HCV was carried out in batches using the Murex (Murex Biotech, South Africa) and Dialab (Dialab, Austria) Elisa kits with 100% sensitivity and 99.87% specificity. Positive samples were repeatedly reactive.

Statistical analysis of data was performed using statistical package for social science (version 9, SPSS) .The chi- squared test was used to determine the association between anti-HCV positivity and other variables. Statistical significance was set at < 0.05.

Results

A total of 334 blood donors were screened for anti-HCV with overall prevalence of 2.1%.

Table i shows that the blood donors comprised 4.5 % (15/334) females and 95.5 % (319/334) males with a M: F ratio of 21.3:1. Although statistically not significant, the overall seroprevalence of anti-HCV among the males was higher than the females. (p=0.05). Voluntary non- remunerated blood donors comprised 5.1 % (17/334) of total blood donors with 0% anti-HCV seroprevalence. Majority of blood donors were family replacement 94.9 % (317/334) with anti-HCV prevalence of 2.2 % (7/317).The difference in anti-HCV positivity between voluntary and family replacement blood donors is not statistically significant (p=0.05)

In table ii, among blood donors with history of multiple sex partners, 6% (6/98) were positive for anti-HCV. Seven (24.1%) of blood donors with history of sexually transmitted diseases were anti-HCV positive. One (5.3%) of blood donors with history of past surgical procedures was positive for anti-HCV. No positive history of illicit intravenous drug use was elicited among blood donors. One (3.2%) of blood donors who had tattoo was positive for anti-HCV. There is significant association between evaluated risk factors and anti-HCV positivity (p<0.05).

Blood donors recruited for this study were aged between 20 and 60years. Although, statistically, there is no significant difference in anti-HCV seroprevalence among age groups (p=0.05), the highest value is in the 20-30years with 3.7%. (Figure 1)

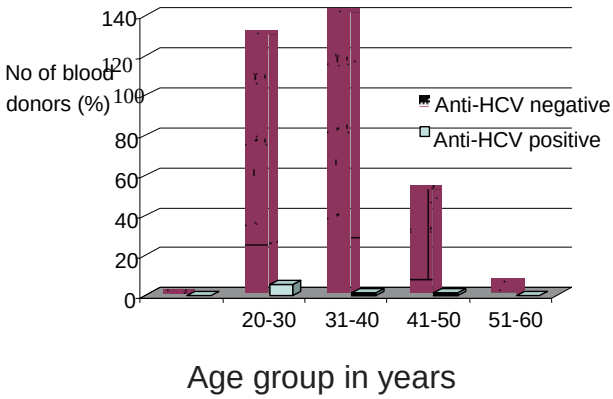
Table i: Demographic data and types of blood donors

Blood donors	Anti-HCV negatives n (%)	Anti-HCV positives n (%)	Total screened n=334	Chi-square P value
Males	312 (97.8)	7 (2.2)	319	X^2=0.319
Females	15 (100)	0	15	P =0.05
M:F ratio	21.3:1	-	-	
Mean age	40	-	-	
Voluntary	17	0	17	X^2=0.00
Family replacement	310	7	317	P =0.05

Table ii: Risk factors and anti-HCV positivity among blood donors

Risk Factors	Anti-HCV negatives n (%)	Anti-HCV positives n (%)	Total blood donors	Chi-square P value
History of Multiple sex partners	92(93.9)	6(6.1)	98	X^2 = 15.9 P< 0.05
History of sexually transmitted diseases	22(75.9)	7(24.1)	29	
Past surgical procedures	18(94.7)	1(5.3)	19	
History of illicit intravenous drug use	-	-	-	
History of Tattooing	30(96.8)	1(3.2)	31	

Age group distribution of anti-HCV among blood donors



Discussion

Our study examined the prevalence of anti-HCV among apparently healthy blood donors, with observed seropositivity rate of 2.1%.

The low prevalence of anti-HCV among blood donors in this study is comparable with reports from Port Harcourt (2.9%), [4] Benin (3%),[5] and Ibadan (3.69%) [6] in Nigeria, Niamey, Niger (3.2%) [7] in Sub- Saharan Africa and the United States of America (2%).[8] Previous studies report low anti-HCV prevalence among pregnant women (1.86%) [9]in Benin and among the general population (3%) [10] in the Niger Delta region of Nigeria.

The observed prevalence value is lower than other reports among blood donors from Benin (12.3%), [11] Oshogbo (6.0%), [12] Jos (4.3%), [13] Ilorin (5.0%), [14] Port Harcourt (5%) [15] in Nigeria and Egypt (9.0%) [16], North Africa. The observed prevalence of anti-HCV among blood donors in this study is however higher than some reported values among the same group in Port Harcourt (0.5%),[17] Kano (0.4%)[18] in Nigeria, Ghana (0.9%),[19] Kenya (0.9%),[20] the Gambia (1.1%),[21] Ethiopia (0.7%)[22] in Africa, North East England (0.18%)[23]and Germany (0.1-1.0%)[8]. The variations in anti-HCV prevalence among blood donors may be due to differences in the sensitivities and specificities of different generations of kits, donor selection criteria and sample size in the various studies.

The observed predominance of male donors is comparable with

those of previous reports in Nigeria.[12, 13, 15] The higher anti-HCV seropositivity among the males with zero prevalence among female donors is at par with a previous report in Nigeria.[17] This finding is at variance with some previous reports[10, 15]

The observed highest prevalence of anti-HCV in age 20-30years is at par with some previous reports [15, 17] though at variance with others[10] in Nigeria.

It is observed in this study, that there is strong association between evaluated risk factors and anti-HCV positivity among blood donors. Of the evaluated risk factors, the highest prevalence (24%) was among subjects with positive history of sexually transmitted diseases $p<0.05$.All anti-HCV positive blood donors had positive history of sexually transmitted diseases. This finding is at par with previous reports.[11] Sexual route may be an important mode of HCV transmission

It was observed, that majority of blood donors were family replacement, three hundred and seventeen (94.9 %) with higher anti-HCV seorpositivity of 2.2%. The voluntary non-remunerated donors were seventeen (5.1%) with anti-HCV seropositivity of 0%. This finding is at par with previous reports in Nigeria.[12, 15, 17]

Reports of HCV prevalence from previous studies among known high risk groups for infection vary widely in Nigeria, ranging from 14% to 47% in HIV positive patients,[24] diabetics,[25] chronic liver disease,[26] sickle cell disease patients,[27] among patients with hepatocellular carcinoma[28] and acute icteric hepatitis.[29]

To improve blood safety and reduce morbidity and mortality from HCV- associated post- transfusion hepatitis and its sequelae, emphasis would be on preventive measures of infection such as donor education. Blood donor selection should be done carefully using

questionnaires to exclude prospective blood donors with high risk behaviours for HCV infection while donated blood units must be screened for HCVmarkers.

Blood donation should be sourced among low risk, voluntary, non-remunerated donors by all blood transfusion services.

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